Report 1: Genomic characterization of the 2026 Bundibugyo virus disease outbreak in the Democratic Republic of the Congo

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Genomic data cutoff: September 6th, 2026; report prepared September 24, 2026

Main findings

- Detection lagged emergence by roughly three months. 786 Bundibugyo virus (BDBV) genomes date the most recent common ancestor of the outbreak to early March 2026 (8 March; 95% HPD 7 February– 30 March), ≈9 weeks before the official declaration of the outbreak on 15 May 2026.
- A single spillover event, then sustained human-to-human transmission. Genetic similarity across all 2026 sequences indicates one introduction from an unknown reservoir, with no evidence of repeated zoonotic events.
- Sub-lineages are not geographically structured within Ituri, implying frequent movement of individuals between health zones. This confirms that human mobility plays an important role in the expansion of the outbreak.
- Genomes from Mongbwalu health zone are deep in the phylogeny and exhibit substantial diversity, consistent with a slow start there followed by export to Bunia and other urban centres and rapid growth from March to June 2026.
- Genomic sequencing would have delivered these answers in near-real time. Re-running the analysis on data available as early as 29 June 2026 (n=121 genomes) reproduces the same growth trajectory — supporting routine phylodynamic outputs as an operational complement to case-based surveillance.
- Early doubling time was estimated to be ~13 days (95% HPDI: 7–23) with a plateauing of infections in Ituri Province in June followed by slow decline since early August. This result should be interpreted in the context of epidemiological data and field observations. The genomic analysis is averaging over the sampled epidemic and at this stage does not account for local heterogeneity in transmission.
- However, from the line list there appears to be an increase in cases in Nord Kivu and future results from the genomic analysis will be able to investigate the epidemic dynamics there.

The latest genomic epidemiology situation and analysis is available on the INRB dashboard: https://inrb-umie.github.io/BDBV2026-Epidemic_Dashboard/genomic-epidemiology.html

Table 1: Key quantitative parameters.

Parameter	Value
Genomes generated/analysed	786 generated; 778 after QC (767 Ituri, 10 Nord Kivu, 1 Sud Kivu)
Median genome coverage	96.8% (IQR 91.2–97.2)
Evolutionary rate	9.8×10^{-4} subs/site/year (95% HPD 8.7–10.8 $\times 10^{-4}$)
Origin of circulating lineages (tMRCA))	8 March 2026 (95% HPD 7 February – 30 March 2026)
Cases, 1 May – 6 Sep 2026	6,550 confirmed; 40.4% test positivity
Sequencing coverage vs. case burden	slope 0.77, $R^2 = 0.73$
Median age, female sex (%)	27 yr (IQR 18–39); 53.3% female

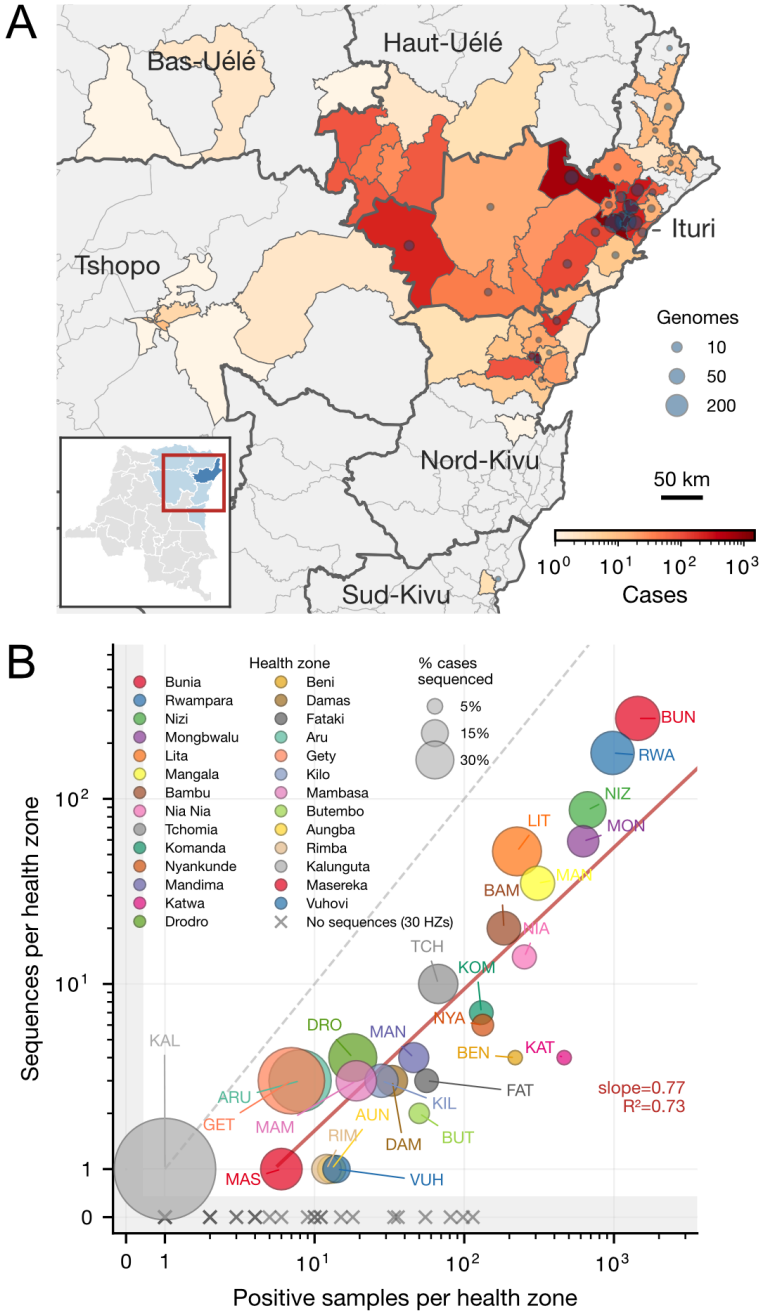


Figure 1. Confirmed cases and genomes from the early 2026 Bundibugyo virus disease outbreak in the DRC. a, Health zones with confirmed cases and genomes through 6 September 2026; health-zone fill shows case counts and overlaid circle size shows genome counts. Inset locates the affected provinces (Ituri in dark blue); the red outline marks the mapped area. b, Genomic sampling intensity by health zone — each circle is a health zone, sized by the percent of cases sequenced. The dashed grey unity line represents one sequence per positive sample; the red line is the regression fit across health zones with ≥1 sequence (slope 0.77, R^2 = 0.73). Health zones with positives but no sequences are marked "×".

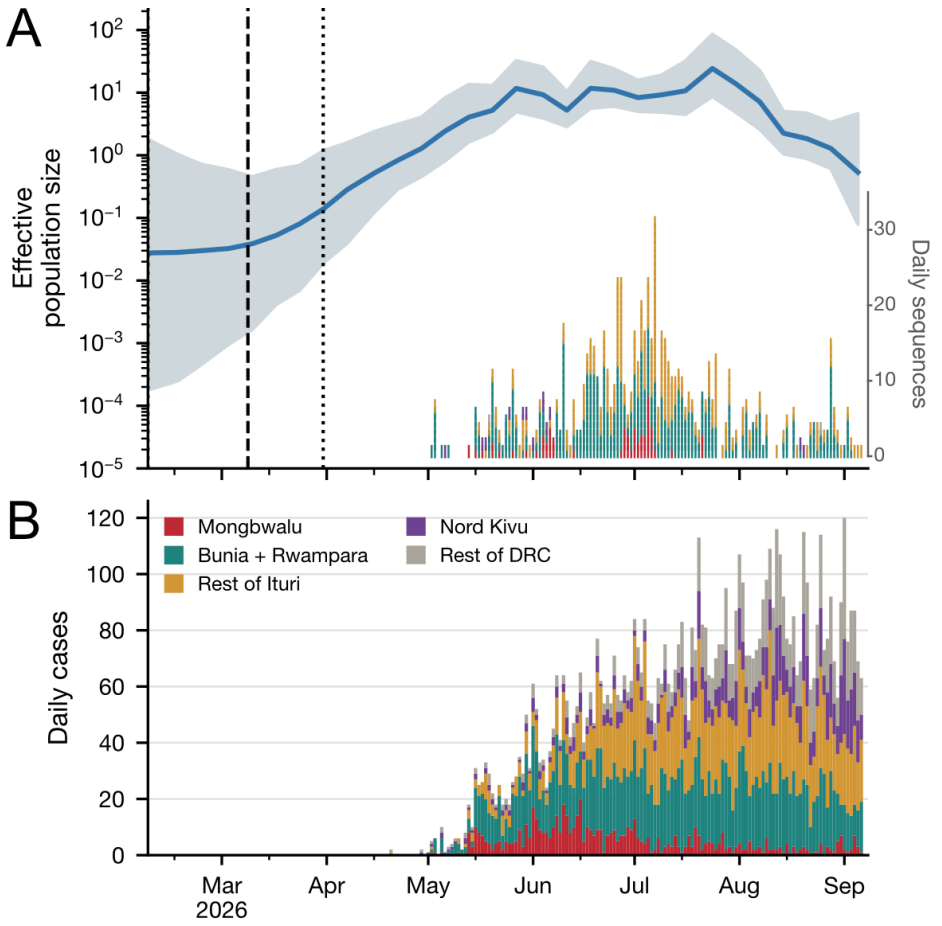


Figure 2. Evolutionary history of the early 2026 BDBV outbreak in the DRC. a, SkyGrid reconstruction of relative virus effective population size, an indirect measure of infections, over time (blue line, mean; shading, 95% HPD). The dashed line marks the median tMRCA and the dotted line the upper bound of its 95% HPD; the lower bound coincides with the y-axis limit. The rug plot shows dates and frequency of included genomes, coloured as in panel b. b, Daily PCR-positive case counts by symptom onset in Mongbwalu (red), Bunia and Rwampara (green), the rest of Ituri (gold), Nord Kivu (purple), and the rest of the DRC (grey).

Implications and value of viral genome sequencing

- Pathogen genomic data and analysis can support assessment of the timing of emergence and spatial transmission dynamics of BDBV in the current outbreak
- Continued effort to sequence samples from health zones newly reporting cases will help put their epidemics in the context of the wider epidemic
- Sampling recent positive cases will allow continued situational awareness (e.g., is the epidemic growing or declining; are clusters of cases in one location related to frequent virus movements or a self-sustaining epidemic)
- Beni and Katwa currently are undersampled with respect to the number of reported cases. They should be prioritised for more sequencing along with Musienene, Pawa, Wamba, and Isiro, each of which has >50 confirmed cases and no sampled genomes

Technical Appendix

1. Data and methods

- Specimens. EDTA whole blood from living patients and oropharyngeal swabs from decedents, collected 1 May – 6 September 2026 in Ituri and Nord Kivu provinces as part of the emergency response (ethics approval ESP/CE/81/2026, Kinshasa School of Public Health).
- Whole genome Sequencing. Samples with Ct < 31 were sequenced using two approaches: early-phase samples used viral probe-based capture enrichment, subsequently by tiled-amplicon using ARTIC pan-Ebola or BDBV-specific primer panels. Sequencing libraries were run on an Illumina NextSeq 2000 or Oxford Nanopore GridION sequencer. Consensus genomes were generated via the ARTIC amplicon-nf pipeline (ONT) or Viralrecon 2.6.0 (Illumina). Median genome coverage 96.8% (IQR 91.2–97.2).
- Quality control. Alignment with MAFFT v7.525, trimmed to 19,120 nt. Three genomes appeared to have sequencing issues and were excluded. Host ADAR-editing signatures (clustered T→C substitutions) were detected with raccoon and masked — including 28 excess T→C sites in the glycoprotein — to prevent inflation of the inferred substitution rate. Iterative root-to-tip regression removed 5 strong temporal outliers, giving a final set of 778 genomes (767 Ituri, 10 Nord Kivu, and 1 Sud Kivu).
- Phylogenetic inference and molecular dating. Maximum-likelihood tree in IQ-TREE 3 (GTR+F+R3, selected by ModelFinder). Time-scaled inference in BEAST X v10.6.0-beta4 using a SkyGrid coalescent model (45 weekly transition points, 46-week cut-off), GTR+I substitution model and a strict molecular clock; 100 million MCMC generations, 30% burn-in, all relevant ESS > 100.
- Retrospective real-time test. The full analysis was repeated on cumulative subsets defined by genome release date — 1 July (n=121), 3 August (n=237), 6 August (n=402) and 10 August 2026 (n=507) — to ask what could have been concluded at each point during the response.
- Epidemiological data. Confirmed-case line list and weekly situation reports, 1 May – 6 September 2026. Missing symptom-onset dates were imputed from the fitted onset-to-collection delay distribution (epidist R package v0.4). A live dashboard is maintained at https://inrb-umie.github.io/BDBV2026-Epidemic_Dashboard/.

2. Additional details on main results

2.1 Epidemiological distribution (Figure 1)

Between 1 May and 6 September 2026, 6,550 PCR-confirmed cases of BVD were reported with an overall test positivity of 40.4%. Ituri province accounted for 5,331 cases (81.4%), concentrated in four health zones: Bunia (1,441; 22.0%), Rwampara (979; 14.9%), Mongbwalu (623; 9.5%) and Nizi (667; 10.2%). Sequencing effort tracked case burden closely but sub-proportionally (slope 0.77, $R^2 = 0.73$ on the log-log scale), and 30 health zones with confirmed cases yielded no sequences at all — a clear geographic gap in genomic coverage.

2.2 Timing of emergence and evolutionary rate (Figure 3)

The estimated evolutionary rate is 9.8×10^{-4} substitutions per site per year (95% HPD 8.7–10.8 $\times 10^{-4}$), consistent with other orthoebolavirus outbreaks. The median tMRCA is 8 March 2026 (95% HPD 7 February – 30 March 2026). Because the tMRCA marks the common ancestor of sampled lineages, the true spillover may be earlier still. We consider it possible that the virus may have been circulating in humans for roughly two to three months before the first laboratory confirmation.

2.3 Transmission dynamics (Figure 2)

The SkyGrid reconstruction shows three phases: a slow start in January–February 2026, marked acceleration from March to June, then a decline in growth rate and stabilisation of effective population size through the end of the study period. The effective population size is an indirect measure of incidence and is mirrored in the case data. Phylogenetic sub-clusters are not geographically structured within Ituri, indicating repeated movement of BDBV-infected individuals between health zones rather than contained local chains. Mongbwalu shows a recent decline in daily case counts.

2.4 Robustness and operational value of real-time genomics

Genome availability fell from late July through early September, raising the possibility that the apparent stabilisation and decline was a sampling artefact. The cumulative-subset analyses argue against this: every cut-off reproduces acceleration from March to June followed by decline and stabilisation. Small end-of-window dips in the earliest subsets are consistent with sparse sampling at the edge rather than genuine change. Once sufficient diversity had accumulated, phylodynamic inference was stable — the key operational point of the paper.

2.5 Viral evolution and mutations

Mutational analysis of 625 sequences identified 558 mutations, most observed in a single genome. Four mutations exceed 10% prevalence, including one non-synonymous substitution in the nucleoprotein (NP) associated with a distinct lineage in the phylogeny. Viral genetic diversity was maintained and increasing throughout the sampling period.

2.6 Diagnostic performance note

On the Altona RT-PCR assay, median Ct was 0.4 cycles lower in oropharyngeal swabs than in whole blood samples (relative difference −1.6%; 95% CI −7.8 to 3.0). On RADIONE, a point of care closed PCR platform, median Ct was 1.4 cycles lower in swabs (−5.3%, 95% CI −10.4 to 0.4), although in both cases the confidence interval included no difference. Higher Ct was modestly associated with lower genome coverage (Altona Spearman $\rho = -0.218$, n=284; RADIONE $\rho = -0.182$, n=361).

3. Limitations and caveats

- Sampling is heavily Ituri-weighted. 767 of 778 analysed genomes are from Ituri, with ten genomes from Nord Kivu and one from Sud Kivu. Conclusions about spread to other provinces are not currently available from the genomic data.
- Estimates of population-size trajectories (Figure 2a) should be considered together with the presented uncertainty. This uncertainty becomes greater towards the present due to the delay between case detection and sequence availability.
- tMRCA is a lower bound on spillover timing. It dates the common ancestor of sampled lineages; unsampled early diversity would push the true introduction earlier.
- Origin attribution is inferential. Mongbwalu-as-origin is supported by phylogenetic depth and diversity plus concordant epidemiological reports, but the reservoir and the spillover event itself remain unidentified.
- Mutation findings are descriptive. No functional, neutralisation or diagnostic-sensitivity testing has been performed on the NP substitution or the other high-frequency mutations.
- Two data sources, two case counts. Figure 1a uses INSP situation-report totals while other analyses use the line list; minor discrepancies reflect differing case-validation timelines.

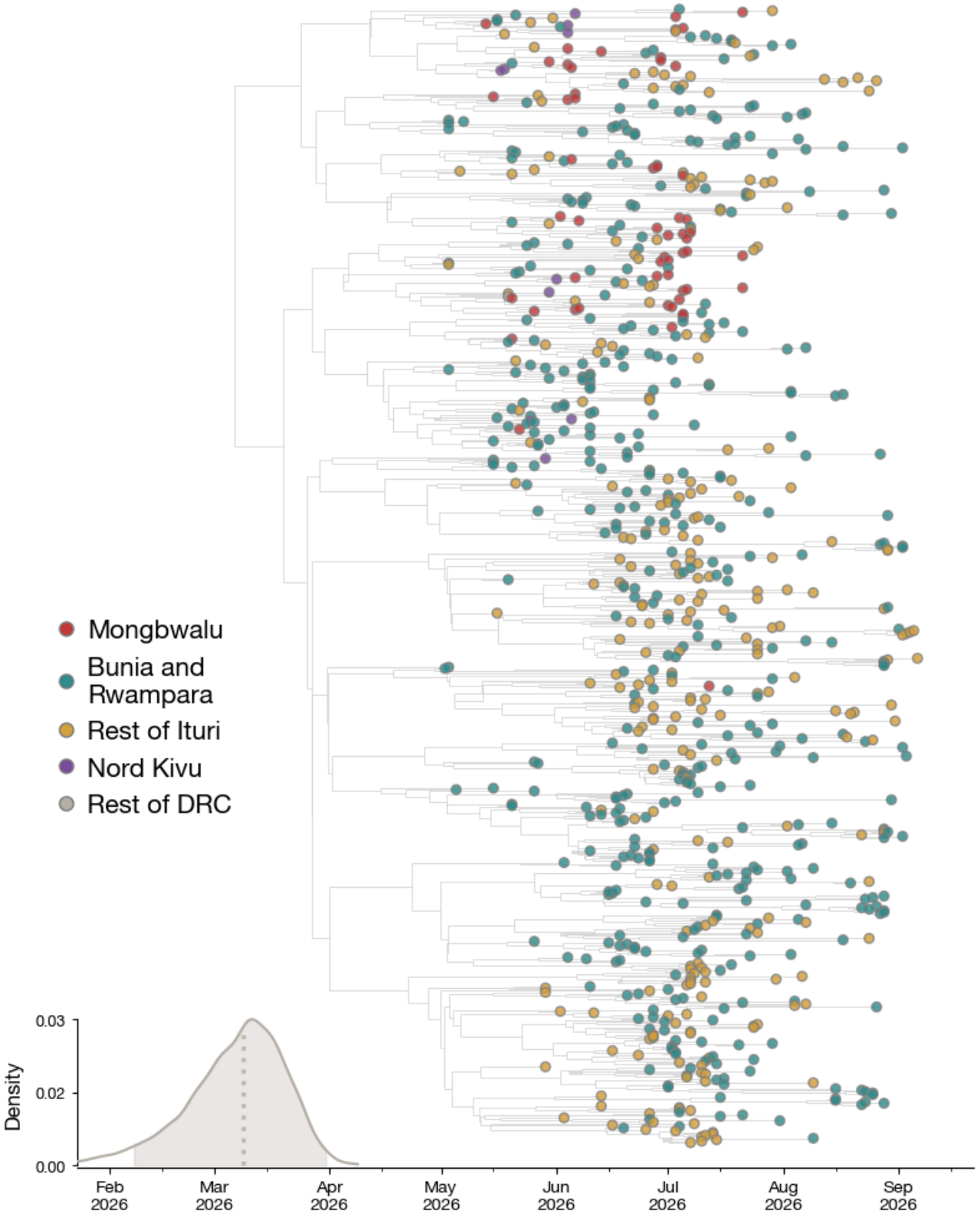


Figure 3. Time-resolved phylogeny of the 2026 BDBV outbreak in the DRC. The density plot shows the time of the most recent common ancestor (tMRCA), with the dotted line indicating the median estimate and the shaded region representing the 95% HPDI.